1 Chapter 13

Problems

13.1 If the light particle (electron) in the Coulomb scattering of Section 13.1 is treated classically, scattering through an angle θ is correlated uniquely to an incident trajectory of impact parameter b according to

$$b = \frac{ze^2}{pv}\cot\frac{\theta}{2}$$

where $p = \gamma mv$ and the differential scattering cross section is $\dfrac{d\sigma}{d\Omega} = \dfrac{b}{\sin\theta}\left|\dfrac{db}{d\theta}\right|$.

(a) Express the invariant momentum transfer squared in terms of impact parameter and show that the energy transfer $T(b)$ is

$$T(b) = \frac{2z^2e^4}{mv^2}\cdot\frac{1}{b^2 + b_{\min}^{(c)2}}$$

where $b_{\min}^{(c)} = ze^2/pv$ and $T(0) = T_{\max} = 2\gamma^2\beta^2mc^2$.

(b) Calculate the small transverse impulse Δp given to the (nearly stationary) light particle by the transverse electric field (11.152) of the heavy particle $q = ze$ as it passes by at large impact parameter b in a (nearly) straight line path at speed v. Find the energy transfer $T \approx (\Delta p)^2/2m$ in terms of b. Compare with the exact classical result of part a. Comment.

Solution.

We use Gaussian units in this chapter, write $S = -dE_{\text{projectile}}/dx > 0$ for a stopping power, and take $e > 0$ as the magnitude of the electron charge. The heavy projectile is treated as infinitely massive in its rest frame. Consequently the magnitude $p = \gamma mv$ of the electron momentum is unchanged in that frame. The stipulated impact-parameter law is the Rutherford model; the word exact below refers to that model, whereas the straight-path impulse is its large-b approximation.

(a) Put $b_c = ze^2/(pv)$. The given relation implies

$$\cot^2\frac{\theta}{2} = \frac{b^2}{b_c^2}, \qquad \sin^2\frac{\theta}{2} = \frac{b_c^2}{b^2 + b_c^2}.$$

The incoming and outgoing four-momenta have equal energy in the projectile rest frame. Hence their invariant difference is

$$Q^2 \equiv -(p_e' - p_e)^2 = |\mathbf{p}_e' - \mathbf{p}_e|^2 = 2p^2(1 - \cos\theta) = \frac{4p^2b_c^2}{b^2 + b_c^2}.$$

Here four-momenta have units of momentum, so Q^2 has units of momentum squared, as in Jackson's (13.2). In the laboratory the electron starts at rest. If it gains kinetic energy T, its final energy and momentum obey

$$E_e' = mc^2 + T, \qquad p_e'^2c^2 = T^2 + 2mc^2T.$$

Therefore the same invariant is

$$Q^2 = p_e'^2 - (T/c)^2 = 2mT.$$

Combining the two evaluations gives

$$\boxed{T(b) = \frac{2p^2 b_c^2}{m(b^2 + b_c^2)} = \frac{2z^2 e^4}{mv^2} \frac{1}{b^2 + b_c^2}.}$$

In particular,

$$T(0) = \frac{2p^2}{m} = 2\gamma^2 mv^2 = 2\gamma^2 \beta^2 mc^2 = T_{\max}.$$

A finite projectile mass changes this maximum by the recoil denominator given immediately after Jackson's (13.4); it is omitted consistently here.

(b) Equation (11.152), with closest approach at $t = 0$, gives

$$E_\perp(t) = \frac{ze\gamma b}{(b^2 + \gamma^2 v^2 t^2)^{3/2}}, \qquad E_\parallel(t) = -\frac{ze\gamma v t}{(b^2 + \gamma^2 v^2 t^2)^{3/2}}.$$

The parallel impulse vanishes by oddness. The magnitude of the transverse impulse, using $u = \gamma v t / b$, is

$$|\Delta p_\perp| = e \int_{-\infty}^{\infty} E_\perp \, dt = \frac{ze^2}{bv} \int_{-\infty}^{\infty} \frac{du}{(1 + u^2)^{3/2}} = \frac{ze^2}{bv} \left[\frac{u}{\sqrt{1 + u^2}} \right]_{-\infty}^{\infty} = \frac{2ze^2}{bv}.$$

The electron's actual impulse points opposite to $\mathbf{E}$; the energy depends on its square. For a nearly stationary, nonrelativistic electron,

$$\boxed{T_{\text{imp}}(b) = \frac{(\Delta p_\perp)^2}{2m} = \frac{2z^2 e^4}{mv^2 b^2}.}$$

Indeed

$$T(b) = T_{\text{imp}}(b) \left[1 - \frac{b_c^2}{b^2} + O\left(\frac{b_c^4}{b^4} \right) \right].$$

Besides $b \gg b_c$, using $(\Delta p)^2/(2m)$ requires $2ze^2/(bvmc) \ll 1$. The cancellation of γ in the impulse is physical: the transverse field is enhanced by γ, but its duration is shortened by the same factor. Extrapolating the impulse result to $b = 0$ would violate its assumptions and miss the finite $T_{\max}$.

13.2 Time-varying electromagnetic fields $\mathbf{E}(\mathbf{x}, t)$ and $\mathbf{B}(\mathbf{x}, t)$ of finite duration act on a charged particle of charge e and mass m bound harmonically to the origin with natural frequency ω_0 and small damping constant Γ. The fields may be caused by a passing charged particle or some other external source. The charge's motion in response to the fields is nonrelativistic and small in amplitude compared to the scale of spatial variation of the fields (dipole approximation). Take the oscillator initially unexcited at the origin, and retain the leading energy transfer in linear response to the applied fields. Show that the energy transferred to the oscillator in the limit of very small damping is

$$\Delta E = \frac{\pi e^2}{m} |\mathbf{E}(\omega_0)|^2$$

where $\mathbf{E}(\omega)$ is the symmetric Fourier transform of $\mathbf{E}(0, t)$:

$$\mathbf{E}(0, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathbf{E}(\omega) e^{-i\omega t} \, d\omega; \qquad \mathbf{E}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathbf{E}(0, t) e^{i\omega t} \, dt$$

Solution.

Work to first order in the applied fields when computing the displacement, and to second order when computing the deposited energy. In the dipole approximation the equation of motion is

$$m(\ddot{\mathbf{x}} + \Gamma\dot{\mathbf{x}} + \omega_0^2\mathbf{x}) = e\mathbf{E}(0,t).$$

The magnetic force is proportional to $\dot{\mathbf{x}} \times \mathbf{B}$, so it changes the first-order motion only at second order in the applied field. It also does no instantaneous work. Spatial gradients of $\mathbf{E}$ are beyond the stated dipole approximation. Assume the oscillator is at the origin with zero velocity before the pulse; an initially excited oscillator would have an additional phase-dependent interference term. The marked assumptions make explicit the scope needed for the formula on printed p. 655; the nonrelativistic and dipole conditions alone do not specify the initial state.

First set $\Gamma = 0$. The retarded solution is

$$\mathbf{x}(t) = \frac{e}{m\omega_0}\int_{-\infty}^{t} \mathbf{E}(0,t')\sin[\omega_0(t-t')]\,dt'.$$

After the pulse, define the real vectors

$$\mathbf{C} = \int_{-\infty}^{\infty} \mathbf{E}(0,t')\cos\omega_0 t'\,dt', \qquad \mathbf{S} = \int_{-\infty}^{\infty} \mathbf{E}(0,t')\sin\omega_0 t'\,dt'.$$

Expansion of the sine gives

$$\mathbf{x} = \frac{e}{m\omega_0}(\mathbf{C}\sin\omega_0 t - \mathbf{S}\cos\omega_0 t), \qquad \dot{\mathbf{x}} = \frac{e}{m}(\mathbf{C}\cos\omega_0 t + \mathbf{S}\sin\omega_0 t).$$

The cross terms cancel between kinetic and potential energies:

$$\Delta E = \frac{m}{2}\left(|\dot{\mathbf{x}}|^2 + \omega_0^2|\mathbf{x}|^2\right) = \frac{e^2}{2m}(|\mathbf{C}|^2 + |\mathbf{S}|^2).$$

By the stated Fourier convention, $\mathbf{E}(\omega_0) = (\mathbf{C} + i\mathbf{S})/\sqrt{2\pi}$. Thus

$$\boxed{\Delta E = \frac{\pi e^2}{m}|\mathbf{E}(\omega_0)|^2.}$$

Even with $\mathbf{B} = 0$, the initial-state condition matters. Let the unforced motion before the pulse be $\mathbf{a}\cos\omega_0 t + \mathbf{b}\sin\omega_0 t$. The pulse changes these two coefficients to $\mathbf{a} - e\mathbf{S}/(m\omega_0)$ and $\mathbf{b} + e\mathbf{C}/(m\omega_0)$, respectively. Subtracting the initial oscillator energy from the final one gives

$$\Delta E = \frac{e^2}{2m}(|\mathbf{C}|^2 + |\mathbf{S}|^2) + e\omega_0(\mathbf{b}\cdot\mathbf{C} - \mathbf{a}\cdot\mathbf{S}).$$

The second term can have either sign and vanishes for an initially unexcited oscillator (or after averaging uniformly over its initial phase).

One can also keep the damping until the final step. Fourier transformation gives

$$\mathbf{x}(\omega) = \frac{e\mathbf{E}(\omega)}{m(\omega_0^2 - \omega^2 - i\Gamma\omega)}.$$

After all the energy has been dissipated, the work done by the external field equals $m\Gamma\int|\dot{\mathbf{x}}|^2 dt$. Parseval's identity and $\mathbf{E}(-\omega) = \mathbf{E}(\omega)^*$ then give

$$W_\Gamma = \frac{2e^2}{m}\int_0^\infty \frac{\Gamma\omega^2|\mathbf{E}(\omega)|^2}{(\omega_0^2 - \omega^2)^2 + \Gamma^2\omega^2}\,d\omega.$$

Near the positive resonance, $\omega = \omega_0 + s$,

$$\frac{2\Gamma\omega^2}{(\omega_0^2 - \omega^2)^2 + \Gamma^2\omega^2} \simeq \frac{\Gamma/2}{s^2 + (\Gamma/2)^2} \longrightarrow \pi\delta(s).$$

This reproduces the boxed result. Thus the small-damping limit can mean either the mechanical energy just after the pulse, before appreciable decay, or the total energy subsequently dissipated.

13.3 The external fields of Problem 13.2 are caused by a charge ze passing the origin in a straight-line path at speed v and impact parameter b. The fields are given by (11.152).

(a) Evaluate the Fourier transforms for the perpendicular and parallel components of the electric field at the origin and show that

$$E_\perp(\omega) = \frac{ze}{bv}\left(\frac{2}{\pi}\right)^{1/2}\xi K_1(\xi); \qquad E_\parallel(\omega) = -i\,\frac{ze}{\gamma bv}\left(\frac{2}{\pi}\right)^{1/2}\xi K_0(\xi)$$

where $\xi = \omega b/\gamma v$, and $K_\nu(\xi)$ is the modified Bessel function of the second kind and order ν. [See references to tables of Fourier transforms in Section 13.3.]

(b) Using the result of Problem 13.2, write down the energy transfer ΔE to a harmonically bound charged particle. From the limiting forms of the modified Bessel functions for small and large argument, show that your result agrees with the appropriate limit of $T(b)$ in Problem 13.1 on the one hand and the arguments at the end of Section 13.1 on the adiabatic behavior for $b \gg \gamma v/\omega_0$ on the other.

Solution.

Take the perpendicular axis from the trajectory toward the observer, so that the fields are those displayed in Problem 13.1(b). We first take $\omega > 0$, as is appropriate when inserting ω_0 in the energy formula.

(a) Set $u = \gamma vt/b$ and $\xi = \omega b/(\gamma v)$. The two transforms become

$$E_\perp(\omega) = \frac{ze}{bv\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{e^{i\xi u}}{(1+u^2)^{3/2}}\,du, \qquad E_\parallel(\omega) = -\frac{ze}{\gamma bv\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{ue^{i\xi u}}{(1+u^2)^{3/2}}\,du.$$

For completeness, the basic transform can be derived from

$$(1+u^2)^{-1/2} = \frac{1}{\sqrt{\pi}}\int_0^\infty s^{-1/2}e^{-s(1+u^2)}\,ds.$$

The Gaussian u integration gives

$$\int_{-\infty}^{\infty}\frac{e^{i\xi u}}{\sqrt{1+u^2}}\,du = \int_0^\infty \frac{ds}{s}e^{-s-\xi^2/(4s)} = 2K_0(\xi).$$

A convergence factor justifies this conditionally convergent transform and can be removed for $\xi > 0$. More generally the denominator $\sqrt{u^2+a^2}$ gives $2K_0(a\xi)$. Differentiating in a and setting $a = 1$ yields

$$\int_{-\infty}^{\infty}\frac{e^{i\xi u}}{(1+u^2)^{3/2}}\,du = 2\xi K_1(\xi).$$

Also,

$$\frac{u}{(1+u^2)^{3/2}} = -\frac{d}{du}(1+u^2)^{-1/2},$$

so integration by parts gives the second integral as $2i\xi K_0(\xi)$. Consequently,

$$\boxed{E_\perp(\omega) = \frac{ze}{bv}\sqrt{\frac{2}{\pi}}\,\xi K_1(\xi), \qquad E_\parallel(\omega) = -i\frac{ze}{\gamma bv}\sqrt{\frac{2}{\pi}}\,\xi K_0(\xi).}$$

For negative frequencies the real-field condition determines the continuation: replace each Bessel argument by $|\xi|$, replace the transverse prefactor ξ by $|\xi|$, and retain the signed ξ in the longitudinal prefactor. This also makes the even/odd Fourier symmetries explicit.

(b) Orthogonality of the parallel and perpendicular directions eliminates cross terms in $|\mathbf{E}|^2$. With $\xi_0 = \omega_0 b/(\gamma v)$, Problem 13.2 gives

$$\boxed{\Delta E(b) = \frac{2z^2e^4}{mv^2b^2}\left[\xi_0^2K_1^2(\xi_0) + \frac{\xi_0^2}{\gamma^2}K_0^2(\xi_0)\right].}$$

At small argument,

$$K_1(\xi) = \xi^{-1} + O(\xi\ln\xi), \qquad K_0(\xi) = -\ln(\xi/2) - \gamma_{\rm E} + O(\xi^2\ln\xi).$$

Thus $\xi^2K_1^2 \to 1$, $\xi^2K_0^2 \to 0$, and

$$\Delta E \longrightarrow \frac{2z^2e^4}{mv^2b^2}.$$

This is the large-b impulse limit of Problem 13.1, with the additional condition $b \ll \gamma v/\omega_0$ that the pulse be sudden on the oscillator time scale. The useful overlap is $b_c \ll b \ll \gamma v/\omega_0$, together with a nonrelativistic response.

For $\xi \gg 1$,

$$K_\nu(\xi) = \sqrt{\frac{\pi}{2\xi}}e^{-\xi}\left[1 + \frac{4\nu^2-1}{8\xi} + O(\xi^{-2})\right],$$

and therefore

$$\Delta E \simeq \frac{\pi z^2e^4}{mv^2b^2}\,\xi_0(1+\gamma^{-2})e^{-2\xi_0}\left[1+O(\xi_0^{-1})\right].$$

For $b \gg \gamma v/\omega_0$ the oscillator follows the slowly varying force and returns almost to its initial state. The remaining excitation is exponentially small, quantitatively establishing adiabatic suppression.

13.4

(a) Taking $\hbar\langle\omega\rangle = 12Z$ eV in the quantum-mechanical energy-loss formula, calculate the rate of energy loss (in MeV/cm) in air at NTP, aluminum, copper, and lead for a proton and a mu meson, each with kinetic energies of 10, 100, 1000 MeV.

(b) Convert your results to energy loss in units of MeV·(cm^2/g) and compare the values obtained in different materials. Explain why, for each fixed projectile and kinetic energy, the energy losses in MeV-(cm^2/g) are within about a factor of 2.3 of each other, whereas the values in MeV/cm differ greatly.

Solution.

(a) Use Jackson's (13.14)–(13.15), with unit projectile charge, exactly as requested:

$$S = \frac{4\pi N Z e^4}{mc^2\beta^2}\left[\ln\frac{2mc^2\beta^2\gamma^2}{I} - \beta^2\right], \qquad I = \hbar\langle\omega\rangle = 12Z \text{ eV}.$$

Here m is the electron mass and $N = \rho N_{\mathrm{A}}/A$ is the atomic number density. For projectile rest energy Mc^2 and kinetic energy T,

$$\gamma = 1 + \frac{T}{Mc^2}, \qquad \beta^2 = 1 - \gamma^{-2}, \qquad \beta^2\gamma^2 = \gamma^2 - 1 = \frac{T(T + 2Mc^2)}{M^2c^4}.$$

Writing $r_0 = e^2/(mc^2)$ for the classical electron radius gives the convenient mass stopping power

$$\boxed{\frac{S}{\rho} = \frac{K}{\beta^2}\frac{Z}{A}\left[\ln\frac{2(0.51099895\text{ MeV})(\gamma^2-1)}{I\text{ (MeV)}} - \beta^2\right], \quad K = 0.30707493\text{ MeV cm}^2\text{ mol}^{-1}.}$$

The coefficient is $K = 4\pi N_{\mathrm{A}} r_0^2 mc^2$, and A is expressed in $\mathrm{g\,mol^{-1}}$. We use $M_p c^2 = 938.27209$ MeV and $M_\mu c^2 = 105.65838$ MeV.

The adopted material parameters are

	air	Al	Cu	Pb
ρ ($\mathrm{g\,cm^{-3}}$)	0.0012931	2.699	8.960	11.35
Z/A ($\mathrm{mol\,g^{-1}}$)	0.499185	0.481811	0.456362	0.395753
I (eV)	87.596	156	348	984

NTP here means 0°C and one atmosphere; this convention matters only for the air column in MeV/cm. The densities and atomic weights are from the Particle Data Group atomic/nuclear-property tables.[1] For air we apply $I_j = 12Z_j$ to each constituent and add their stopping powers. This is equivalent to

$$\left\langle\frac{Z}{A}\right\rangle = \sum_j w_j\frac{Z_j}{A_j}, \qquad \ln I_{\text{air}} = \frac{\sum_j w_j(Z_j/A_j)\ln(12Z_j\text{ eV})}{\sum_j w_j Z_j/A_j}.$$

The dry-air mass fractions for C, N, O, Ar are respectively 0.000124, 0.755267, 0.231781, 0.012827, normalized to sum to one. Thus the supplied $12Z$ prescription is maintained for the mixture instead of inserting the tabulated experimental excitation energy.

For example, a 10 MeV proton has $\gamma = 1.0106579$, $\beta^2 = 0.0209798$, and $2mc^2(\gamma^2-1) = 0.0219008$ MeV. For aluminum the substitution is

$$\frac{S}{\rho} = \frac{0.30707493(13/26.9815385)}{0.0209798}\left[\ln\frac{0.0219008}{0.000156} - 0.0209798\right] = 34.7207\text{ MeV cm}^2/\text{g},$$

and $S = 2.699(34.7207) = 93.7112$ MeV/cm. Applying the same calculation to every case gives

		S (MeV/cm)			
particle	T (MeV)	air	Al	Cu	Pb
p	10	0.051969	93.711	246.646	202.605
p	100	0.008311	15.485	43.198	39.634
p	1000	0.002533	4.801	13.781	13.266
μ	10	0.009082	16.904	47.064	43.035
μ	100	0.002600	4.924	14.116	13.558
μ	1000	0.002613	5.031	14.802	14.813

[1] PDG, 2005 Table 6.1 gives the air density at 0°C. The dry-air table gives mass fractions; the aluminum, copper, and lead tables give elemental densities and molar masses.

These are the model's local collisional stopping powers, not an average over a slab in which the particle slows appreciably. Density, shell, radiative, and finite-projectile-mass corrections have not been added to the explicitly requested (13.14).

(b) Dividing each column by its density produces

		S/ρ (MeV cm^2/g)			
particle	T (MeV)	air	Al	Cu	Pb
p	10	40.189	34.721	27.527	17.851
p	100	6.427	5.737	4.821	3.492
p	1000	1.959	1.779	1.538	1.169
μ	10	7.024	6.263	5.253	3.792
μ	100	2.011	1.825	1.575	1.195
μ	1000	2.020	1.864	1.652	1.305

At a fixed projectile speed the material dependence is through Z/A, which changes only from about 0.50 to 0.40, and through a logarithm of I. The density factor, which differs by almost four orders of magnitude between air and lead, has disappeared. This explains the similarity of mass stopping powers and the large differences in MeV/cm.

The original assertion of a factor strictly less than two has one exception within the problem's own model:

$$\left.\frac{(S/\rho)_{\text{air}}}{(S/\rho)_{\text{Pb}}}\right|_{p,\ 10\,\text{MeV}} = \frac{40.1894}{17.8506} = 2.25143.$$

The other five row-wise ratios lie between 1.55 and 1.85. Thus about a factor of two, or the marked bound of 2.3, is accurate; a factor of two for all 24 entries irrespective of projectile energy would also be incorrect.

13.5 Consider the energy loss by close collisions of a fast, but nonrelativistic, heavy particle of charge ze passing through an electronic plasma. Assume that the screened Coulomb interaction $V(r) = ze^2 \exp(-k_{\text{D}} r)/r$, where k_{D} is the Debye screening parameter, acts between the electrons and the incident particle.

(a) Show that the energy transfer in a collision at impact parameter b is given approximately by

$$\Delta E(b) \simeq \frac{2(ze^2)^2}{mv^2}\, k_{\text{D}}^2 K_1^2(k_{\text{D}} b)$$

where m is the electron mass and v is the velocity of the incident particle.

(b) Determine the energy loss per unit distance traveled for collisions with impact parameter greater than $b_{\min}$. Assuming $k_{\text{D}} b_{\min} \ll 1$, show that

$$\left(\frac{dE}{dx}\right)_{k_{\text{D}} b<1} \simeq \frac{(ze)^2}{v^2}\, \omega_p^2 \ln\left(\frac{1}{1.47 k_{\text{D}} b_{\min}}\right)$$

where $b_{\min}$ is given by the larger of the classical and quantum minimum impact parameters [(13.16) and above].

Solution.

(a) During a weak collision use the undeflected trajectory $r(t) = \sqrt{b^2 + v^2 t^2}$. Integrating the interaction along that path gives

$$\mathcal{V}(b) = \int_{-\infty}^{\infty} V(r(t))\, dt = \frac{ze^2}{v} \int_{-\infty}^{\infty} \frac{e^{-k_{\text{D}}\sqrt{b^2+z^2}}}{\sqrt{b^2+z^2}}\, dz.$$

Set $z = b\sinh u$, so that $r = b\cosh u$ and $dz/r = du$. The integral representation $K_0(x) = \int_0^\infty e^{-x\cosh u}\,du$ gives

$$\mathcal{V}(b) = \frac{2ze^2}{v} K_0(k_{\mathrm{D}} b).$$

The transverse force is $-\partial V/\partial b$, and differentiation commutes with the convergent integral. Therefore

$$|\Delta p_\perp| = \left| -\frac{d\mathcal{V}}{db} \right| = \frac{2ze^2 k_{\mathrm{D}}}{v} K_1(k_{\mathrm{D}} b).$$

The longitudinal force is odd in t and integrates to zero. With a nonrelativistic electron recoil,

$$\boxed{\Delta E(b) \simeq \frac{(\Delta p_\perp)^2}{2m} = \frac{2(ze^2)^2}{mv^2} k_{\mathrm{D}}^2 K_1^2(k_{\mathrm{D}} b).}$$

For $k_{\mathrm{D}} b \ll 1$ the factor $k_{\mathrm{D}} K_1(k_{\mathrm{D}} b)$ tends to $1/b$, recovering the unscreened impulse.

(b) Let n_e be the electron density, so $\omega_p^2 = 4\pi n_e e^2/m$. The number of electrons encountered per unit path length in an annulus is $2\pi n_e b\,db$. Consequently, putting $u = k_{\mathrm{D}} b_{\min}$,

$$S_{\mathrm{close}} = 2\pi n_e \int_{b_{\min}}^\infty b\,\Delta E(b)\,db = \frac{(ze)^2 \omega_p^2}{v^2} \int_u^\infty x K_1^2(x)\,dx.$$

The upper limit may be infinite because screening suppresses the large-b integrand exponentially. To evaluate the last integral, use

$$K_0' = -K_1, \qquad K_1' = -K_0 - K_1/x, \qquad K_2 = K_0 + 2K_1/x.$$

Differentiating the expression below with these identities gives $-uK_1^2(u)$, and it vanishes at infinity. Hence

$$\int_u^\infty x K_1^2(x)\,dx = \frac{u^2}{2}\left[K_0(u)K_2(u) - K_1^2(u)\right] = uK_0K_1 + \frac{u^2}{2}(K_0^2 - K_1^2).$$

As $u \to 0$, $uK_1 \to 1$, $u^2 K_0^2 \to 0$, and $K_0 = -\ln(u/2) - \gamma_{\mathrm{E}} + o(1)$, giving

$$\int_u^\infty x K_1^2(x)\,dx = \ln\frac{2}{u} - \gamma_{\mathrm{E}} - \frac{1}{2} + O(u^2 \ln^2 u) = \ln\frac{1}{C_{\mathrm{c}} u} + O(u^2 \ln^2 u), \quad C_{\mathrm{c}} = \frac{e^{\gamma_{\mathrm{E}}+1/2}}{2} = 1.468\ldots.$$

Thus

$$\boxed{S_{\mathrm{close}} \simeq \frac{(ze)^2\omega_p^2}{v^2} \ln\frac{1}{1.47\,k_{\mathrm{D}} b_{\min}}.}$$

The labels close and $k_{\mathrm{D}} b < 1$ refer to the dominant range; the screened integral itself has not been sharply truncated at $k_{\mathrm{D}} b = 1$. In this nonrelativistic, heavy-projectile approximation the cutoff is

$$b_{\min} \sim \max\left(\frac{ze^2}{mv^2}, \frac{\hbar}{mv}\right).$$

It excludes either large classical deflections or distances below the electron's reduced de Broglie wavelength. The logarithmic formula requires $k_{\mathrm{D}} b_{\min} \ll 1$ and a speed well above the electron thermal speed.

13.6 The energy loss in a plasma from distant collisions can be found with Fermi's method for the density effect. Consider the nonrelativistic limit of (13.36) with the relative dielectric constant of a plasma given by (7.59) augmented by some damping,

$$\epsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\Gamma}$$

Assume that the arguments of the Bessel functions are small (corresponding to a speed of the incident particle large compared to thermal speeds in the plasma).

(a) Show that the energy loss (13.36) for $k_{\rm D}b > 1$ becomes

$$\left(\frac{dE}{dx}\right)_{k_{\rm D}b>1} \approx \frac{2z^2e^2}{\pi v^2}\int_0^\infty \mathrm{Re}\left(\frac{i\omega}{\epsilon(\omega)}\right)\ln\left(\frac{1.123k_{\rm D}v}{\omega}\right)d\omega$$

(b) With the assumption that $\Gamma \ll \omega_p$ in $\epsilon(\omega)$, show that the formula of part a yields

$$\left(\frac{dE}{dx}\right)_{k_{\rm D}b>1} \approx \frac{z^2e^2}{v^2}\,\omega_p^2\ln\left(\frac{1.123k_{\rm D}v}{\omega_p}\right)$$

Combine with the close-collision result of Problem 13.5 to find the total energy loss of a nonrelativistic particle passing through a plasma,

$$\left(\frac{dE}{dx}\right) \approx \frac{z^2e^2}{v^2}\,\omega_p^2\ln\left(\frac{\Lambda v}{\omega_p b_{\rm min}}\right)$$

where Λ is a number of order unity. The presence of ω_p in the logarithm suggests that the energy loss may be quantized in units of $\hbar\omega_p$. In fact, electrons passing through thin metallic foils do show this discreteness in energy loss, allowing determination of effective plasma frequencies in metals. [See H. Raether, *Springer Tracts in Modern Physics*, Vol. 38, ed. G. Höhler, Springer-Verlag, Berlin (1965), pp. 84–157.]

Solution.

(a) With a cylindrical separation radius a, Jackson's (13.36), on printed p. 634, is

$$S_{b>a} = \frac{2z^2e^2}{\pi v^2}\,\mathrm{Re}\int_0^\infty i\omega\,\lambda^* aK_1(\lambda^* a)K_0(\lambda a)\left(\frac{1}{\epsilon} - \beta^2\right)d\omega, \qquad \lambda^2 = \frac{\omega^2}{v^2}(1-\beta^2\epsilon).$$

For $\beta \ll 1$ set $\lambda = \omega/v$ to leading order and drop β^2. Put $a = k_{\rm D}^{-1}$. At the frequencies responsible for energy deposition, $u = \omega/(k_{\rm D}v) \ll 1$, and

$$uK_1(u) = 1 + O(u^2\ln u), \qquad K_0(u) = \ln(2/u) - \gamma_{\rm E} + O(u^2\ln u).$$

The Bessel factors are then real and

$$\boxed{S_{b>k_{\rm D}^{-1}} \simeq \frac{2z^2e^2}{\pi v^2}\int_0^\infty \mathrm{Re}\left(\frac{i\omega}{\epsilon(\omega)}\right)\ln\frac{2e^{-\gamma_{\rm E}}k_{\rm D}v}{\omega}\,d\omega.}$$

Since $2e^{-\gamma_{\rm E}} = 1.12291897$, this is the stated 1.123.

(b) Algebraically,

$$\frac{1}{\epsilon} = \frac{\omega^2 + i\Gamma\omega}{\omega^2 - \omega_p^2 + i\Gamma\omega} = 1 + \frac{\omega_p^2}{\omega^2 - \omega_p^2 + i\Gamma\omega}.$$

Multiplication by the conjugate denominator therefore gives the positive loss function

$$\operatorname{Re}\frac{i\omega}{\epsilon} = \frac{\Gamma\omega_p^2\omega^2}{(\omega^2-\omega_p^2)^2+\Gamma^2\omega^2}.$$

For $\omega = \omega_p + s$ with $|s|, \Gamma \ll \omega_p$, this becomes

$$\frac{\Gamma\omega_p^2}{4s^2+\Gamma^2} = \frac{\omega_p^2}{2}\frac{\Gamma/2}{s^2+(\Gamma/2)^2} \longrightarrow \frac{\pi\omega_p^2}{2}\delta(\omega-\omega_p).$$

The logarithm varies slowly across this narrow resonance. Its value at ω_p can be taken outside the integral, yielding

$$\boxed{S_{\text{distant}} \simeq \frac{z^2e^2\omega_p^2}{v^2}\ln\frac{1.12291897\,k_{\text{D}}v}{\omega_p}.}$$

The small-argument condition used above is therefore specifically $\omega_p/(k_{\text{D}}v) \ll 1$. For a thermal Debye plasma ω_p/k_{D} is its electron thermal-speed scale.

Adding Problem 13.5 makes the arbitrary matching scale k_{D} cancel inside the logarithm:

$$S = S_{\text{close}} + S_{\text{distant}} = \frac{z^2e^2\omega_p^2}{v^2}\ln\left[\frac{2e^{-\gamma_{\text{E}}}}{C_{\text{c}}}\,\frac{v}{\omega_p b_{\min}}\right].$$

Thus, retaining the unrounded constants,

$$\boxed{S \simeq \frac{z^2e^2\omega_p^2}{v^2}\ln\frac{\Lambda v}{\omega_p b_{\min}}, \qquad \Lambda = 4e^{-2\gamma_{\text{E}}-1/2} = 0.764803\ldots.}$$

Using 1.123 and 1.47 literally instead gives 0.76395, a difference from rounding. The occurrence of ω_p here arises from the collective pole in $1/\epsilon$; classical theory establishes that frequency, while the stated discrete losses $\hbar\omega_p$ require quantization of the collective excitation.

13.7 With the same approximations as were used to discuss multiple scattering, show that the *projected* transverse displacement y (see Fig. 13.7) of an incident particle is described approximately by a Gaussian distribution,

$$P(y)\,dy = A\,\exp\left[\frac{-y^2}{2\langle y^2\rangle}\right]dy$$

where the mean square displacement is $\langle y^2\rangle = (x^2/6)\langle\Theta^2\rangle$, x being the thickness of the material traversed and $\langle\Theta^2\rangle$ the mean square angle of scattering.

Solution.

Let s measure distance from the entrance and let $\delta\theta_y$ be the projected angular kick in a collision at s. In the small-angle approximation this kick contributes $(x-s)\delta\theta_y$ to the final displacement. With independent, azimuthally symmetric collisions,

$$y = \sum_j (x-s_j)\delta\theta_{yj}, \qquad \langle\delta\theta_{yj}\rangle = 0, \qquad \langle\delta\theta_{yj}^2\rangle = \frac{1}{2}\langle\theta^2\rangle.$$

Write $N\sigma\,ds$ for the mean number of collisions in ds. The mean projected angular variance added per unit length is

$$D_\theta = N\sigma\,\frac{\langle\theta^2\rangle}{2}.$$

All cross terms in $\langle y^2 \rangle$ vanish because kicks at distinct positions are independent with zero mean. Thus

$$\langle y^2 \rangle = D_\theta \int_0^x (x-s)^2 \, ds = D_\theta \left[-\frac{(x-s)^3}{3} \right]_0^x = \frac{D_\theta x^3}{3}.$$

The total, unprojected exit-angle variance is

$$\langle \Theta^2 \rangle = N\sigma x \langle \theta^2 \rangle = 2 D_\theta x.$$

Eliminating D_θ gives

$$\boxed{\langle y^2 \rangle = \frac{x^2}{6} \langle \Theta^2 \rangle.}$$

To establish the Gaussian shape rather than just its variance, take the characteristic function of the sum:

$$\langle e^{iky} \rangle = \prod_j \left\langle e^{ik(x-s_j)\delta\theta_{yj}} \right\rangle.$$

Keeping the second cumulant in the same many-small-kicks approximation used for the angular distribution,

$$\ln \langle e^{iky} \rangle \simeq -\frac{k^2}{2} \sum_j (x-s_j)^2 \langle \delta\theta_{yj}^2 \rangle = -\frac{k^2}{2} \langle y^2 \rangle.$$

The inverse Fourier transform is a normalized Gaussian:

$$\boxed{P(y) = \frac{1}{\sqrt{2\pi \langle y^2 \rangle}} \exp\left[-\frac{y^2}{2\langle y^2 \rangle} \right].}$$

This is the central part of the distribution. Rare, unusually large single kicks cause non-Gaussian tails, just as in the angular problem. Constant D_θ also assumes negligible slowing over x; if the momentum changes, the general variance is $\int_0^x D_\theta(s)(x-s)^2 \, ds$ instead.

13.8 If the finite size of the nucleus is taken into account in the "single-scattering" tail of the multiple-scattering distribution, there is a critical thickness x_c beyond which the single-scattering tail is effectively suppressed within the angular-cutoff approximation.

(a) Define x_c in a reasonable way and calculate its value (in cm) for aluminum and lead, assuming that the incident particle is relativistic and singly charged.

(b) For these thicknesses calculate the number of collisions that occur and determine whether the Gaussian approximation is valid.

Solution.

(a) A sharp disappearance is a feature of an effective angular cutoff, not of the exact nuclear form factor. We define a conservative critical thickness by the criterion suggested at the end of Section 13.6:

$$\boxed{\langle \Theta^2 \rangle (x_c) = \theta_{\max}^2.}$$

At this thickness the typical cumulative deflection is already as large as the largest effective individual deflection, so a distinct single-scattering tail cannot extend outside the central distribution.

Use the source's estimates, on printed pp. 641–644,

$$a = 1.4a_0 Z^{-1/3}, \qquad R = 1.4A^{1/3}\ \text{fm}, \qquad \theta_{\min} = \frac{\hbar}{pa}, \qquad \theta_{\max} = \frac{\hbar}{pR}, \qquad L = \ln\frac{\theta_{\max}}{\theta_{\min}} = \ln\frac{a}{R}.$$

For a singly charged relativistic projectile ($z = 1$, $v \simeq c$), (13.65) is

$$\langle\Theta^2\rangle = 2\pi N x \left(\frac{2Ze^2}{pc}\right)^2 L.$$

Equating this to $\hbar^2/(p^2R^2)$ cancels the incident momentum:

$$\frac{\hbar^2}{p^2R^2} = \frac{8\pi N Z^2 e^4 L}{p^2c^2} x_c, \qquad \boxed{x_c = \frac{1}{8\pi N Z^2 \alpha^2 R^2 L}.}$$

For charge ze and speed βc, the corresponding expression has an additional factor β^2/z^2. The assumed relativistic singly charged case is the one tabulated here.

Use $a_0 = 5.2917721 \times 10^{-9}$ cm and $N = \rho N_{\text{A}}/A$, with the material data of Problem 13.4:

	Al	Pb
Z	13	82
A	26.9815	207.2
N (cm^{-3})	6.024×10^{22}	3.299×10^{22}
a (cm)	3.15076×10^{-9}	1.70526×10^{-9}
R (cm)	4.19904×10^{-13}	8.28434×10^{-13}
L	8.92313	7.62969
x_c (cm)	46.65	0.6433

These are estimates within the source's screening and nuclear-radius model, not precision material thresholds.

The definition is not unique. In Fig. 13.8 the tail starts near $\theta_{\text{proj}} \simeq 2.5\sqrt{\langle\Theta^2\rangle}$, because the dimensionless projected angle is $\alpha_{\text{ang}} = \theta_{\text{proj}}/\sqrt{\langle\Theta^2\rangle}$. Equating that crossover to $\theta_{\max}$ instead gives the earlier onset of strong tail suppression at $x_c/2.5^2 \simeq 7.46$ cm in aluminum and 0.103 cm in lead. Both definitions express the same comparison of cumulative and single-event deflections. The adopted boxed definition specifies which one is used for the requested numbers.

(b) The total cross section (13.59) and mean number of collisions are

$$\sigma = \pi\left(\frac{2Ze^2}{pc}\right)^2 \theta_{\min}^{-2} = \pi a^2 (2Z\alpha)^2, \qquad n_c = N\sigma x_c.$$

Substitution of the expression for x_c cancels N, $Z^2\alpha^2$:

$$\boxed{n_c = \frac{a^2}{2R^2L} = \begin{cases} 3.15 \times 10^6, & \text{Al}, \\ 2.78 \times 10^5, & \text{Pb}. \end{cases}}$$

Equivalently, $\langle\Theta^2\rangle = n_c\langle\theta^2\rangle$ and $\langle\theta^2\rangle = 2\theta_{\min}^2 L$ give the same result. Even the earlier crossover definition gives about 5.05×10^5 and 4.44×10^4 collisions. Both are much larger than the $n \lesssim 200$ regime identified in the source as problematic for a Gaussian approximation. The many-collision approximation is therefore well justified. The incident energy must also be large enough that slowing and large total deflections over these thicknesses can be neglected; relativistic speed by itself does not guarantee negligible energy loss through 47 cm of aluminum.

13.9 Assuming that Plexiglas or Lucite has an index of refraction of 1.50 in the visible region, compute the angle of emission of visible Cherenkov radiation for electrons and protons as a function of their kinetic energies in MeV. Determine how many quanta with wavelengths between 4000 and 6000 Å are emitted per centimeter of path in Lucite by a 1 MeV electron, a 500 MeV proton, and a 5 GeV proton.

Solution.

For refractive index $n = 1.50$, (13.50) gives $\cos\theta_C = 1/(n\beta)$. The velocity at kinetic energy T is

$$\beta(T) = \frac{\sqrt{T(T + 2Mc^2)}}{T + Mc^2}.$$

The requested functions of energy, with T and Mc^2 in MeV, are therefore

$$\boxed{\theta_C(T; M) = \cos^{-1}\left[\frac{T + Mc^2}{1.50\sqrt{T(T + 2Mc^2)}}\right], \qquad T > T_{\text{th}}.}$$

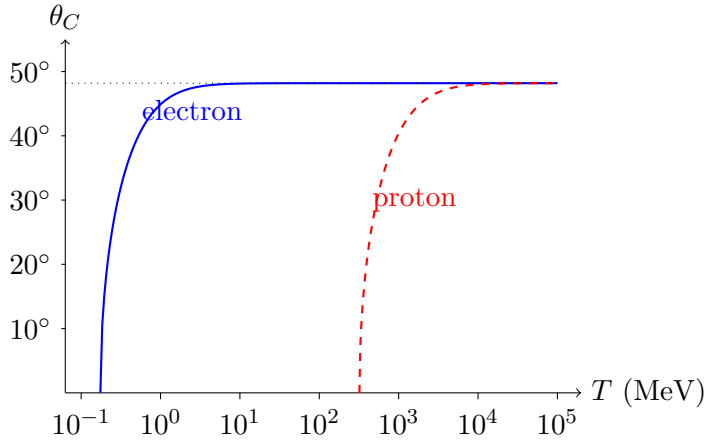
There is no Cherenkov emission below threshold. Setting $n\beta = 1$ gives

$$\gamma_{\text{th}} = \frac{1}{\sqrt{1 - n^{-2}}} = \frac{3}{\sqrt{5}}, \qquad T_{\text{th}} = Mc^2\left(\frac{3}{\sqrt{5}} - 1\right).$$

With the masses used in Problem 13.4,

$$T_{\text{th},e} = 0.174578 \text{ MeV}, \qquad T_{\text{th},p} = 320.552 \text{ MeV}.$$

Both curves approach the common limit $\theta_C(\infty) = \cos^{-1}(2/3) = 48.1897°$:



The horizontal energy axis is logarithmic; each curve starts at its own threshold.

To count photons, divide the Frank–Tamm spectral energy (13.48) by $\hbar\omega$:

$$\frac{d^2N_\gamma}{dx\,d\omega} = \frac{z^2e^2}{\hbar c^2}\left(1 - \frac{1}{n^2\beta^2}\right).$$

The specified wavelengths are vacuum wavelengths, so $\omega = 2\pi c/\lambda$ and $|d\omega/d\lambda| = 2\pi c/\lambda^2$. Thus

$$\frac{d^2N_\gamma}{dx\,d\lambda} = \frac{2\pi z^2\alpha}{\lambda^2}\left(1 - \frac{1}{n^2\beta^2}\right),$$

and for singly charged particles

$$\frac{N_\gamma}{x} = 2\pi\alpha\left(\frac{1}{4.000\times10^{-5}\text{ cm}} - \frac{1}{6.000\times10^{-5}\text{ cm}}\right)\left(1-\frac{1}{2.25\beta^2}\right)$$
$$= \boxed{382.088\left(1-\frac{1}{2.25\beta^2}\right)\text{cm}^{-1}.}$$

The resulting angles and yields are

particle and kinetic energy	β	θ_C	N_γ/x (cm^{-1})
e, 1 MeV	0.94108	44.8946°	190.34
p, 500 MeV	0.75791	28.4042°	86.46
p, 5000 MeV	0.98744	47.5344°	207.92

These are emitted photon numbers. Absorption, interface escape losses, and detector efficiency would be additional experimental factors.

13.10 A particle of charge ze moves along the z axis with constant speed v, passing $z=0$ at $t=0$. The medium through which the particle moves is described by a dielectric constant $\epsilon(\omega)$.

(a) Beginning with the potential $\Phi(\mathbf{k},\omega)$ of (13.25), show that the potential of frequency ω is given as a function of spatial coordinate $\mathbf{x}$ by

$$\Phi(\omega,\mathbf{x}) = \frac{ze}{v\epsilon(\omega)}\sqrt{\frac{2}{\pi}}\,K_0\left(\frac{|\omega|\,\rho}{v}\sqrt{1-\beta^2\epsilon}\right)e^{i\omega z/v}$$

where z and $\rho=\sqrt{x^2+y^2}$ are the cylindrical coordinates of the observation point.

(b) Assuming that ϵ is independent of frequency and that $\beta^2\epsilon<1$, take the Fourier transform with respect to ω of the expression in part a and obtain $\Phi(\mathbf{x},t)$. Calculate the electric and magnetic fields and compare them to the vacuum fields (11.152). Show that, among other things, the vacuum factor γ is replaced by $\Gamma=(1-\beta^2\epsilon)^{-1/2}$.

(c) Repeat the calculations of parts a and b with $\beta^2\epsilon>1$. Show that now

$$\Phi(\omega,\mathbf{x}) = \frac{ze}{v\epsilon(\omega)}\sqrt{\frac{\pi}{2}}\,e^{i\omega z/v}\left[-N_0\left(\frac{|\omega|\,\rho}{v}\sqrt{\beta^2\epsilon-1}\right)\pm iJ_0\left(\frac{|\omega|\,\rho}{v}\sqrt{\beta^2\epsilon-1}\right)\right]$$

for $\omega\gtrless0$. Calculate the remaining Fourier transform to obtain $\Phi(\mathbf{x},t)$. Relate your answer to the result given in Section 13.4 for $\mathbf{A}(\mathbf{x},t)$.

Solution.

Write $q=ze$, let ρ denote cylindrical radius, and assume a nonmagnetic medium. The retarded prescription is essential, especially above the Cherenkov threshold: take $\operatorname{Im}\epsilon(\omega)>0$ for $\omega>0$ and then approach the lossless limit.

(a) The symmetric four-dimensional transform in (13.25) is

$$\Phi(\mathbf{k},\omega) = \frac{2q}{\epsilon(\omega)}\frac{\delta(\omega-vk_z)}{k_\perp^2+k_z^2-\epsilon(\omega)\omega^2/c^2}.$$

Performing the k_z integral in the three-dimensional inverse transform gives

$$\Phi(\omega,\mathbf{x}) = \frac{2q\,e^{i\omega z/v}}{(2\pi)^{3/2}v\epsilon}\int \frac{d^2k_\perp\, e^{i\mathbf{k}_\perp\cdot\boldsymbol{\rho}}}{k_\perp^2+\lambda^2}, \qquad \lambda^2 = \frac{\omega^2}{v^2}(1-\beta^2\epsilon).$$

The azimuthal integral is $2\pi J_0(k_\perp\rho)$. Initially take $\lambda > 0$ real, so that the auxiliary exponential integral converges. A useful evaluation of the radial integral is

$$\begin{aligned}\int_0^\infty \frac{kJ_0(k\rho)}{k^2+\lambda^2}\,dk &= \int_0^\infty e^{-s\lambda^2}ds\int_0^\infty kJ_0(k\rho)e^{-sk^2}\,dk \\ &= \frac{1}{2}\int_0^\infty \frac{ds}{s}e^{-s\lambda^2-\rho^2/(4s)} = K_0(\lambda\rho).\end{aligned}$$

Analytic continuation to $\mathrm{Re}\,\lambda > 0$ then implements the retarded prescription. The intermediate real-s representation requires $\mathrm{Re}(\lambda^2) > 0$; the continued Bessel identity has the wider domain. Combining the normalization factors gives

$$\boxed{\Phi(\omega,\mathbf{x}) = \frac{q}{v\epsilon(\omega)}\sqrt{\frac{2}{\pi}}\,K_0\!\left(\frac{|\omega|\rho}{v}\sqrt{1-\beta^2\epsilon(\omega)}\right)e^{i\omega z/v}.}$$

The square-root branch is continued from positive real part, and the negative-frequency potential is the complex conjugate of the positive-frequency potential. Equation (13.25) also gives, at each frequency, $\mathbf{A} = \beta\epsilon\Phi\,\hat{\mathbf{z}}$.

(b) Now let ϵ be a real positive constant and $\beta^2\epsilon < 1$. Define

$$\zeta = vt - z, \qquad \Gamma = (1-\beta^2\epsilon)^{-1/2}, \qquad a = \frac{\rho}{\Gamma v}, \qquad \tau = \frac{\zeta}{v}.$$

Pairing positive and negative frequencies gives

$$\Phi(\mathbf{x},t) = \frac{2q}{\pi v\epsilon}\int_0^\infty K_0(a\omega)\cos(\tau\omega)\,d\omega.$$

For clarity, using $K_0(a\omega) = \int_0^\infty e^{-a\omega\cosh u}du$ evaluates the integral as

$$\begin{aligned}\int_0^\infty K_0(a\omega)\cos(\tau\omega)d\omega &= \int_0^\infty \frac{a\cosh u}{a^2\cosh^2 u+\tau^2}\,du \\ &= a\int_0^\infty \frac{ds}{a^2s^2+a^2+\tau^2} = \frac{\pi}{2\sqrt{a^2+\tau^2}},\end{aligned}$$

where $s = \sinh u$. Consequently

$$\boxed{\Phi(\mathbf{x},t) = \frac{q}{\epsilon\sqrt{\zeta^2+\rho^2/\Gamma^2}} = \frac{q\Gamma}{\epsilon\sqrt{\rho^2+\Gamma^2\zeta^2}}, \qquad A_z = \beta\epsilon\Phi.}$$

Set $D = \rho^2+\Gamma^2\zeta^2$. Direct differentiation yields

$$E_\rho = -\partial_\rho\Phi = \frac{q\Gamma\rho}{\epsilon D^{3/2}}.$$

For the longitudinal field, $\partial_z = -\partial_\zeta$ and $\partial_t = v\partial_\zeta$, so both potentials contribute:

$$\begin{aligned}E_z &= -\partial_z\Phi - \frac{1}{c}\partial_t A_z = (1-\beta^2\epsilon)\partial_\zeta\Phi \\ &= -\Gamma^{-2}\frac{q\Gamma^3\zeta}{\epsilon D^{3/2}} = -\frac{q\Gamma\zeta}{\epsilon D^{3/2}}.\end{aligned}$$

Finally,

$$\boxed{\mathbf{E} = \frac{q\Gamma}{\epsilon[\rho^2 + \Gamma^2(z-vt)^2]^{3/2}}[\rho\,\hat{\boldsymbol{\rho}} + (z-vt)\hat{\mathbf{z}}], \qquad \mathbf{B} = \beta\epsilon E_\rho\,\hat{\boldsymbol{\phi}}.}$$

For $\epsilon = 1$, $\Gamma = \gamma$, and these are precisely the uniform-motion vacuum fields (11.152). In a medium the electric field also carries a factor $1/\epsilon$, and the magnetic-to-transverse-electric ratio becomes $\beta\epsilon$.

(c) Let $\kappa = \sqrt{\beta^2\epsilon - 1}$ and now put $a = \kappa\rho/v$. The retarded continuation gives $\lambda\rho = -ia\omega$ for $\omega > 0$. The Bessel continuation

$$K_0(-ix) = \frac{\pi}{2}[-N_0(x) + iJ_0(x)], \qquad x > 0,$$

and its complex conjugate for $\omega < 0$ give

$$\boxed{\Phi(\omega, \mathbf{x}) = \frac{q}{v\epsilon}\sqrt{\frac{\pi}{2}}\,e^{i\omega z/v}[-N_0(a|\omega|) + i\ \mathrm{sgn}(\omega)J_0(a|\omega|)].}$$

Here N_0 is Jackson's Neumann function. Pairing the two frequency halves now yields

$$\Phi(\mathbf{x}, t) = \frac{q}{v\epsilon}\int_0^\infty [-N_0(a\omega)\cos(\tau\omega) + J_0(a\omega)\sin(\tau\omega)]\,d\omega.$$

These are retarded, regulated integrals. Their values follow by setting $s = 0^+ - i\tau$ in the Laplace transforms

$$\int_0^\infty e^{-s\omega}J_0(a\omega)d\omega = \frac{1}{\sqrt{a^2+s^2}}, \qquad \int_0^\infty e^{-s\omega}N_0(a\omega)d\omega = -\frac{2}{\pi}\frac{\mathrm{arsinh}(s/a)}{\sqrt{a^2+s^2}}.$$

Specifically,

$$\int_0^\infty J_0(a\omega)\sin(\tau\omega)d\omega = \frac{\mathrm{sgn}\,\tau\,H(|\tau|-a)}{\sqrt{\tau^2-a^2}}, \qquad \int_0^\infty N_0(a\omega)\cos(\tau\omega)d\omega = -\frac{H(|\tau|-a)}{\sqrt{\tau^2-a^2}}.$$

They cancel for $\tau < -a$, both have zero relevant part for $|\tau| < a$, and add for $\tau > a$. Thus

$$\boxed{\Phi(\mathbf{x}, t) = \frac{2q}{\epsilon}\frac{H(\zeta - \kappa\rho)}{\sqrt{\zeta^2 - \kappa^2\rho^2}}, \qquad \mathbf{A} = \frac{2q\beta\,H(\zeta-\kappa\rho)}{\sqrt{\zeta^2-\kappa^2\rho^2}}\hat{\mathbf{z}}.}$$

The vector potential agrees with (13.51), after rotating Jackson's direction of motion from x to z. It is nonzero only inside the wake behind the particle. The factor of two reflects the two retarded source positions contributing inside a superluminal cone.

For points strictly inside the cone, put $S = \zeta^2 - \kappa^2\rho^2$. The regular parts of the fields are

$$\boxed{E_\rho = -\frac{2q\kappa^2\rho}{\epsilon S^{3/2}}, \qquad E_z = \frac{2q\kappa^2\zeta}{\epsilon S^{3/2}}, \qquad B_\phi = \beta\epsilon E_\rho.}$$

They vanish strictly outside the cone. At the front one must differentiate the whole regulated potential, including its support, before removing the regulator; the divergent front is not represented by these interior formulas alone. The assumption of a frequency-independent dielectric constant creates this singularity, as discussed immediately after (13.51). Dispersion and absorption regularize a physical front.

13.11 A magnetic monopole with magnetic charge g passes through matter and loses energy by collisions with electrons, just as does a particle with electric charge ze.

(a) In the same approximation as presented in Section 13.1, show that the energy loss per unit distance is given approximately by (13.14), but with $ze \to \beta g$ and with only the leading logarithm retained, yielding

$$\left(\frac{dE}{dx}\right)_{\substack{\text{magnetic}\\ \text{monopole}}} \simeq 4\pi NZ\,\frac{g^2e^2}{mc^2}\ln\!\left(\frac{2\gamma^2mv^2}{\hbar\langle\omega\rangle}\right)$$

(b) With the Dirac quantization condition (6.153) determining the magnetic charge, what z value is necessary for an ordinary charged particle in order that it lose energy at relativistic speeds at the same rate as a monopole? Sketch for the magnetic monopole a curve of dE/dx equivalent to Fig. 13.1 and comment on the differences.

Solution.

(a) In Gaussian units the static monopole field is $\mathbf{B}' = g\mathbf{r}'/r'^3$. A boost along its path gives

$$B_\perp(t) = \frac{g\gamma b}{(b^2+\gamma^2v^2t^2)^{3/2}}, \qquad \mathbf{E} = -\boldsymbol{\beta}\times\mathbf{B}.$$

An initially slow electron is accelerated by this transverse electric field. Its magnitude is exactly the electric-charge expression of Problem 13.1(b) with ze replaced by βg, although its direction is rotated by 90° around the trajectory. Thus

$$|\Delta p| = \frac{2e\beta g}{bv} = \frac{2eg}{bc}, \qquad \Delta E(b) = \frac{2e^2g^2}{mc^2b^2}.$$

For electron density $n_e = NZ$, integrating over impact parameters gives

$$S_g = 2\pi n_e\int b\,\Delta E(b)\,db = \frac{4\pi NZg^2e^2}{mc^2}\ln\frac{b_{\max}}{b_{\min}}.$$

The bound-electron and quantum matching used in Section 13.1 sets the logarithm to $\ln B_q$, with $B_q = 2\gamma^2mv^2/(\hbar\langle\omega\rangle)$, to the accuracy retained here. Therefore

$$\boxed{S_g \simeq \frac{4\pi NZg^2e^2}{mc^2}\ln\frac{2\gamma^2mv^2}{\hbar\langle\omega\rangle}.}$$

There is no $1/\beta^2$ prefactor: the extra β in the induced electric field cancels it.

The displayed result is a *leading-logarithm* formula. A literal algebraic substitution in the full electric-charge (13.14) would leave $\ln B_q - \beta^2$ rather than $\ln B_q$. The omitted nonlogarithmic term, and monopole-specific corrections beyond this impulse treatment, cannot be inferred from the replacement rule. The marked qualification in the statement makes the approximation consistent.

(b) Converting the Dirac condition (6.153) to the Gaussian normalization used here gives

$$eg = \frac{n\hbar c}{2}, \qquad \frac{g}{e} = \frac{n}{2\alpha}.$$

Since $\beta \simeq 1$ at relativistic speed, the equivalent ordinary charge is

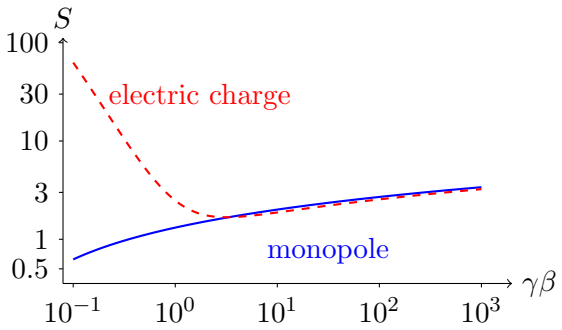
$$\boxed{z_{\text{equivalent}} \simeq \frac{|g|}{e} = 68.518\,|n|.}$$

For the minimum monopole, the stopping power at such speeds is therefore roughly $(68.518)^2 \simeq 4695$ times that of a singly charged particle, at the same leading-logarithm accuracy.

To compare shapes with Fig. 13.1, put $u = \gamma\beta$ and use the figure's $I = 160$ eV. Divide out the squared effective charge:

$$S_{\text{plot,g}} = 0.15 \ln \frac{2mc^2u^2}{I}, \qquad S_{\text{plot,e}} = \frac{0.15}{\beta^2}\left[\ln \frac{2mc^2u^2}{I} - \beta^2\right], \qquad \beta^2 = \frac{u^2}{1+u^2}.$$

The resulting sketch uses logarithmic axes:



Its ordinate S is the same approximate aluminum scale as Fig. 13.1, after dividing the physical stopping power by $(g/e)^2$ for the monopole or by z^2 for the electric particle. The monopole curve rises logarithmically with u, whereas the electric curve has a large low-speed rise and a minimum. Both rise logarithmically at large u. In particular, the monopole lacks the electric particle's $1/\beta^2$ increase near the end of its range. Neither this high-speed approximation nor a negative extrapolation of its logarithm describes the actual stopping law arbitrarily close to rest.

13.12 A relativistic particle of charge ze moves along the z axis with a constant speed βc. The half-space $z \le 0$ is filled with a uniform isotropic dielectric medium with plasma frequency ω_1, and the space $z > 0$ with a similar medium whose plasma frequency is ω_2. Discuss the emission of transition radiation as the particle traverses the interface, using the approximation of Section 13.7.

(a) Show that the radiation intensity per unit circular frequency interval and per unit solid angle is given approximately by

$$\frac{d^2I}{d\omega\, d\Omega} \simeq \frac{z^2e^2\theta^2}{\pi^2 c}\left|\frac{1}{\dfrac{1}{\gamma^2}+\dfrac{\omega_1^2}{\omega^2}+\theta^2} - \frac{1}{\dfrac{1}{\gamma^2}+\dfrac{\omega_2^2}{\omega^2}+\theta^2}\right|^2$$

where θ is the angle of emission relative to the velocity of the particle and $\gamma = (1-\beta^2)^{-1/2}$.

(b) Show that the total energy radiated is

$$I \simeq \frac{z^2e^2}{3c} \cdot \frac{(\omega_1-\omega_2)^2}{(\omega_1+\omega_2)} \cdot \gamma$$

Solution.

(a) Adopt the assumptions of Section 13.7: $\gamma \gg 1$, $\theta \ll 1$, $\omega \gg \omega_j$, negligible absorption and reflection, and $\epsilon_j = 1 - \omega_j^2/\omega^2$. Define

$$A = \gamma^{-2} + \theta^2, \qquad q_j = \frac{\omega}{v} - k_j \cos\theta \simeq \frac{\omega}{2c}\left(A + \frac{\omega_j^2}{\omega^2}\right).$$

The polarization driving the radiation is proportional to $\omega_j^2 \mathbf{E}_i$, while its longitudinal phase is $e^{iq_j z}$. With convergence factors at infinity the two half-space integrals have opposite signs:

$$\int_{-\infty}^{0} e^{iq_1 z + 0^+ z} dz = \frac{1}{iq_1}, \qquad \int_{0}^{\infty} e^{iq_2 z - 0^+ z} dz = -\frac{1}{iq_2}.$$

The transverse integral and its normalization can be kept explicit. Write $q = ze$, $\ell = \omega/(\gamma v)$, $h = k \sin\theta$, and $C = \sqrt{2/\pi}\, q\omega/(\gamma v^2)$. The incident fields are $E_\rho = CK_1(\ell\rho)$ and $E_z = -iCK_0(\ell\rho)/\gamma$ after removing their common z phase. The two-dimensional transform and its derivative give

$$\int d^2\rho\, e^{-ihx} K_0(\ell\rho) = \frac{2\pi}{\ell^2 + h^2}, \qquad \int d^2\rho\, e^{-ihx} \cos\phi K_1(\ell\rho) = -\frac{2\pi i h}{\ell(\ell^2 + h^2)}.$$

For the second identity use $\cos\phi K_1(\ell\rho) = -\ell^{-1}\partial_x K_0(\ell\rho)$ and integrate by parts. Projection onto the radiation polarization in the observation plane gives

$$T = \cos\theta\, T_\rho - \sin\theta\, T_z = -\frac{2\pi i C}{\ell^2 + h^2}\left(\frac{h\cos\theta}{\ell} - \frac{\sin\theta}{\gamma}\right) \simeq -\frac{2i\sqrt{2\pi}\, q}{\omega}\frac{\theta}{A}.$$

Here $v \simeq c$, $h \simeq \omega\theta/c$, and the retained denominator is $\ell^2 + h^2 \simeq (\omega^2/c^2)A$. Generalizing (13.79) to the two media therefore gives

$$\frac{d^2 I}{d\omega\, d\Omega} = \frac{1}{32\pi^3 c^3}\left| T\left(\frac{\omega_1^2}{iq_1} - \frac{\omega_2^2}{iq_2}\right)\right|^2.$$

Inserting q_j leaves a common amplitude factor $4\sqrt{2\pi}\, qc$ multiplying

$$\begin{aligned}\frac{\theta}{A}\left[\frac{\omega_2^2}{A\omega^2 + \omega_2^2} - \frac{\omega_1^2}{A\omega^2 + \omega_1^2}\right] &= \frac{\theta}{A}\left[\left(1 - \frac{A\omega^2}{A\omega^2 + \omega_2^2}\right) - \left(1 - \frac{A\omega^2}{A\omega^2 + \omega_1^2}\right)\right] \\ &= \theta\left[\frac{1}{A + \omega_1^2/\omega^2} - \frac{1}{A + \omega_2^2/\omega^2}\right].\end{aligned}$$

Squaring and using $(4\sqrt{2\pi}\, qc)^2/(32\pi^3 c^3) = q^2/(\pi^2 c)$ gives

$$\boxed{\frac{d^2 I}{d\omega\, d\Omega} = \frac{z^2 e^2 \theta^2}{\pi^2 c}\left|\frac{1}{\gamma^{-2} + \theta^2 + \omega_1^2/\omega^2} - \frac{1}{\gamma^{-2} + \theta^2 + \omega_2^2/\omega^2}\right|^2.}$$

The two sides contribute coherently; adding their intensities separately would miss the cancellation when $\omega_1 = \omega_2$.

(b) Put $a = \omega_1$, $b = \omega_2$. For real, lossless media, the squared difference is

$$\left[\frac{1}{A + a^2/\omega^2} - \frac{1}{A + b^2/\omega^2}\right]^2 = \frac{(a^2 - b^2)^2 \omega^4}{(A\omega^2 + a^2)^2 (A\omega^2 + b^2)^2}.$$

We need its frequency integral. One convenient derivation starts with the partial-fraction integral

$$J(a, b) = \int_0^\infty \frac{s^2\, ds}{(s^2 + a^2)(s^2 + b^2)} = \frac{\pi}{2(a + b)}.$$

Indeed, its integrand is $[b^2/(s^2 + b^2) - a^2/(s^2 + a^2)]/(b^2 - a^2)$. Differentiating J gives

$$\begin{aligned}\int_0^\infty \frac{s^4\, ds}{(s^2 + a^2)^2 (s^2 + b^2)^2} &= -\frac{1}{2a}\partial_a J - \frac{b}{4a}\partial_a \partial_b J \\ &= \frac{\pi}{4a(a + b)^2} - \frac{\pi b}{4a(a + b)^3} = \frac{\pi}{4(a + b)^3}.\end{aligned}$$

The continuous limit covers $a = b$ or one vanishing frequency. Setting $s = \sqrt{A}\,\omega$ therefore gives

$$\int_0^\infty d\omega \left[\frac{1}{A + a^2/\omega^2} - \frac{1}{A + b^2/\omega^2}\right]^2 = \frac{\pi(a-b)^2}{4A^{5/2}(a+b)}.$$

Now $d\Omega \simeq 2\pi\theta\, d\theta$, so

$$I = \frac{z^2e^2}{2c}\frac{(a-b)^2}{a+b}\int_0^\infty \frac{\theta^3\, d\theta}{(\theta^2+\gamma^{-2})^{5/2}}.$$

To show the last step explicitly, set $u = \theta^2 + \gamma^{-2}$:

$$\begin{aligned}\int_0^\infty \frac{\theta^3\, d\theta}{(\theta^2+\gamma^{-2})^{5/2}} &= \frac{1}{2}\int_{\gamma^{-2}}^\infty (u - \gamma^{-2})u^{-5/2}\, du \\ &= \frac{1}{2}\left[-2u^{-1/2} + \frac{2}{3}\gamma^{-2}u^{-3/2}\right]_{\gamma^{-2}}^\infty = \frac{2\gamma}{3}.\end{aligned}$$

Hence

$$\boxed{I \simeq \frac{z^2e^2}{3c}\frac{(\omega_1-\omega_2)^2}{\omega_1+\omega_2}\,\gamma.}$$

Equal media give zero; setting one plasma frequency to zero recovers (13.87). Extending the high-frequency, small-angle approximation over the full integration range gives the leading term proportional to γ; low-frequency details of a real dielectric affect subleading terms.

13.13 Consider the transition radiation emitted by a relativistic particle traversing a dielectric foil of thickness a perpendicular to its path. Assuming that reflections can be ignored because

$$|[n(\omega) - 1]/[n(\omega) + 1]|$$

is very small, show that the differential angular and frequency spectrum is given by the single-interface result (13.84) times the factor,

$$\mathcal{F} = 4\sin^2\Theta, \qquad \text{with } \Theta = \nu\left(1 + \frac{1}{\nu^2} + \eta\right)\frac{a}{4D}$$

Here $D = \gamma c/\omega_p$ is the formation length, $\nu = \omega/\gamma\omega_p$, and $\eta = (\gamma\theta)^2$. Provided $a \gg D$, the factor $\mathcal{F}$ oscillates extremely rapidly in angle or frequency, averaging to $\langle\mathcal{F}\rangle = 2$. For such foils the smoothed intensity distribution is just twice that for a single interface. Frequency distributions for different values of $\Gamma = 2D/a$ are displayed in Fig. 1 of G. B. Yodh, X. Artru, and R. Ramaty, *Astrophys. J.* **181**, 725 (1973).

Solution.

Use the same forward-radiation and high-frequency approximation as in Section 13.7. At fixed frequency and observation angle, the polarization source at depth z carries the driving phase $e^{i\omega z/v}$ and the radiation propagator carries $e^{-ikz\cos\theta}$. Define

$$q_f = \frac{\omega}{v} - k\cos\theta.$$

The entire change from a semi-infinite medium to a foil is then in the longitudinal integral. A half-space, with a vanishing convergence factor, has integral i/q_f, whereas a foil from 0 to a has

$$\int_0^a e^{iq_f z}dz = \frac{e^{iq_f a} - 1}{iq_f} = \frac{2e^{iq_f a/2}}{q_f}\sin\frac{q_f a}{2}.$$

Their squared ratio is

$$\boxed{\mathcal{F} = |1 - e^{iq_f a}|^2 = 2 - 2\cos(q_f a) = 4\sin^2\frac{q_f a}{2}.}$$

This is also the interference of the two interface amplitudes. The entrance and exit discontinuities have opposite sign, and their phase difference is $q_f a$.

To express the phase in the requested variables, expand each factor:

$$\beta^{-1} = 1 + \frac{1}{2\gamma^2} + O(\gamma^{-4}), \qquad n = 1 - \frac{\omega_p^2}{2\omega^2} + O(\omega_p^4/\omega^4), \qquad \cos\theta = 1 - \frac{\theta^2}{2} + O(\theta^4).$$

Consequently,

$$q_f = \frac{\omega}{c}(\beta^{-1} - n\cos\theta) \simeq \frac{\omega}{2c}\left(\gamma^{-2} + \frac{\omega_p^2}{\omega^2} + \theta^2\right).$$

With $\nu = \omega/(\gamma\omega_p)$, $\eta = \gamma^2\theta^2$, and $D = \gamma c/\omega_p$,

$$\boxed{\Theta = \frac{q_f a}{2} = \frac{\nu a}{4D}(1 + \nu^{-2} + \eta), \qquad \mathcal{F} = 4\sin^2\Theta.}$$

For a very thin foil, $q_f a \ll 1$, the factor is $\mathcal{F} \simeq (q_f a)^2$ and the two interfaces almost cancel. For a thick foil the phase oscillates through many cycles within an ordinary frequency or angular bin. Phase averaging gives explicitly

$$\langle\mathcal{F}\rangle = \frac{1}{\pi}\int_{\Theta_0}^{\Theta_0+\pi} 4\sin^2\Theta\, d\Theta = 2.$$

Thus the smoothed thick-foil intensity is twice the single-interface intensity. The criterion is loss of phase resolution, not a pointwise limit of $\sin^2\Theta$ as $a \to \infty$.

13.14 Transition radiation is emitted by a relativistic particle traversing normally a uniform array of N dielectric foils, each of thickness a, separated by air gaps (effectively vacuum), each of length b. Assume that multiple reflections can be neglected for the whole stack. This requires

$$\left|\frac{n(\omega) - 1}{n(\omega) + 1}\right| \simeq \frac{\omega_p^2}{4\omega^2} \ll \frac{1}{N}$$

(a) Show that if the dielectric constant of the medium varies in the z direction as $\epsilon(\omega, z) = 1 - (\omega_p^2/\omega^2)\rho(z)$, the differential spectrum of transition radiation is given approximately by the single-interface result (13.84) times

$$\mathcal{F} = \left|\mu\int dz\ \rho(z)e^{i\omega z/v}\exp\left(-i\cos\theta\int^z k(z')\,dz'\right)\right|^2$$

where $\rho(0) = 1$ by convention, $\mu = \omega/v - k(0)\cos\theta$, and $k(z) = (\omega/c)\sqrt{\epsilon(\omega, z)}$.

(b) Show that for the stack of N foils

$$\mathcal{F} = 4\sin^2\Theta\,\frac{\sin^2[N(\Theta + \Psi)]}{\sin^2[\Theta + \Psi]}$$

where Θ is defined in Problem 13.13 and $\Psi = \nu(1 + \eta)(b/4D)$. Compare G. M. Garibyan, *Zh. Eksp. Teor. Fiz.* **60**, 39 (1970) [transl. *Sov. Phys. JETP* **33**, 23 (1971)].

The practical theory of multilayered transition radiation detectors is treated in great detail by X. Artru, G. B. Yodh, and G. Mennessier, *Phys. Rev. D* **12**, 1289 (1975).

Solution.

(a) In the weak-susceptibility approximation used in (13.76),

$$\mathbf{P}(\mathbf{x},\omega) = -\frac{\omega_p^2}{4\pi\omega^2}\rho(z)\,\mathbf{E}_i(\mathbf{x},\omega).$$

The exciting field has z dependence $e^{i\omega z/v}$. Neglecting reflected waves, propagation from a source at z adds the phase $\exp[-i\cos\theta\int^z k(z')dz']$. Corrections to the transverse radiation kernel from the small change of index are discarded in amplitude, while these accumulated phases are retained. Define

$$\varphi(z) = \frac{\omega z}{v} - \cos\theta\int^z k(z')\,dz', \qquad \varphi'(z) = \frac{\omega}{v} - k(z)\cos\theta.$$

The transverse integral in (13.79) is common to all z. The new longitudinal amplitude is therefore proportional to

$$L[\rho] = \int dz\,\rho(z)e^{i\varphi(z)}.$$

For the reference half-space with $\rho = 1$ and $k = k(0)$, the same integral is

$$L_{\text{ref}} = \lim_{\delta\downarrow 0}\int_0^\infty e^{i\mu z-\delta z}\,dz = \frac{i}{\mu}, \qquad \mu = \frac{\omega}{v} - k(0)\cos\theta.$$

Taking the ratio of squared amplitudes gives

$$\boxed{\mathcal{F} = \left|\mu\int dz\,\rho(z)e^{i\omega z/v}\exp\left[-i\cos\theta\int^z k(z')\,dz'\right]\right|^2.}$$

A change in the lower limit of the phase integral gives an overall phase and does not affect $\mathcal{F}$. The convention $\rho(0) = 1$ sets the reference single-interface normalization. This formula is an approximation for propagation in weakly refracting material; for strongly varying indices the transverse propagation and reflections cannot in general be represented by this phase alone.

(b) Number the foils $j = 0,\ldots,N-1$ and place the jth entrance at $z_j = j(a+b)$. Let

$$q_f = \frac{\omega}{v} - k_f\cos\theta, \qquad q_g = \frac{\omega}{v} - \frac{\omega}{c}\cos\theta, \qquad 2\Theta = q_f a, \quad 2\Psi = q_g b.$$

At $z = z_j + s$ inside foil j the accumulated phase is

$$\varphi(z) = j(q_f a + q_g b) + q_f s, \qquad 0 < s < a.$$

The gaps have $\rho = 0$ and contribute only to phase, not to the source integral. Since $\mu = q_f$, direct integration yields

$$\begin{aligned}\mu L[\rho] &= q_f\sum_{j=0}^{N-1} e^{ij(q_f a+q_g b)}\int_0^a e^{iq_f s}\,ds \\ &= \frac{e^{2i\Theta}-1}{i}\sum_{j=0}^{N-1} e^{2ij(\Theta+\Psi)}.\end{aligned}$$

The finite geometric series, with $\chi = \Theta + \Psi$, is

$$\sum_{j=0}^{N-1} e^{2ij\chi} = \frac{1-e^{2iN\chi}}{1-e^{2i\chi}} = e^{i(N-1)\chi}\frac{\sin(N\chi)}{\sin\chi}.$$

The squared modulus gives

$$\boxed{\mathcal{F} = 4\sin^2\Theta\,\frac{\sin^2[N(\Theta+\Psi)]}{\sin^2(\Theta+\Psi)}.}$$

The vacuum gap phase is, to the same order as in Problem 13.13,

$$q_g \simeq \frac{\omega}{2c}(\gamma^{-2}+\theta^2), \qquad \boxed{\Psi = \frac{\nu b}{4D}(1+\eta).}$$

At $\chi = \ell\pi$ the apparent $0/0$ is removable: the original series equals N, up to its common phase, and the array factor is N^2. At $N = 1$ this reduces to the foil factor. At $b = 0$ the product reduces to $4\sin^2(N\Theta)$, exactly the result for a single contiguous slab of thickness Na.

Garibyan's cited paper, printed p. 24, Eq. (5), has the same foil sine factor and squared array factor. Its Eq. (3) requires $|r|N \ll 1$, not merely a small reflection coefficient at one surface, in agreement with the condition in this problem.[2] The finite-N expression here retains the peak widths, of order $1/N$ in χ. Replacing these peaks by delta functions needs the remaining spectral and angular factors to vary little over those widths; no such additional approximation is needed for the displayed answer.

13.15

(a) Find the number N_γ of transition radiation quanta with frequencies greater than ω_p emitted per interface, starting from the energy spectrum (13.85). Show that for $\gamma \gg 1$,

$$N_\gamma = \frac{z^2e^2}{\pi\hbar c}\left[(\ln\gamma - 1)^2 + \frac{\pi^2}{12}\right]$$

where terms of order $\gamma^{-2}\ln\gamma$ have been neglected.

(b) Using the result from part a for the number of photons and the value $\hbar\omega_p = 20$ eV, find the mean energy of the photons (in keV) for $\gamma = 10^3$, 10^4, 10^5.

Solution.

(a) Write the checked source spectrum (13.85) as

$$\frac{dI}{d\nu} = \frac{z^2e^2\gamma\omega_p}{\pi c}f(\nu), \qquad f(\nu) = (1+2\nu^2)\ln(1+\nu^{-2}) - 2, \qquad \nu = \frac{\omega}{\gamma\omega_p}.$$

A photon has energy $\hbar\omega = \hbar\gamma\omega_p\nu$. The cutoff $\omega > \omega_p$ is therefore $\nu > 1/\gamma$, giving

$$N_\gamma = \frac{z^2\alpha}{\pi}\mathcal{N}(\gamma), \qquad \mathcal{N}(\gamma) = \int_{1/\gamma}^{\infty}\frac{f(\nu)}{\nu}\,d\nu.$$

We evaluate the integral rather than integrate only the low-frequency asymptote, which would lose the constant term. With $y = \nu^{-2}$, $d\nu/\nu = -dy/(2y)$, so

$$\mathcal{N}(\gamma) = \frac{1}{2}\int_0^{\gamma^2}\left[\frac{\ln(1+y)}{y} + \frac{2\ln(1+y)}{y^2} - \frac{2}{y}\right]dy.$$

[2] G. M. Garibyan, *Sov. Phys. JETP* **33**, 23–29 (1971), publisher PDF, p. 24, Eqs. (3)–(5).

The combination of the last two terms has a finite lower limit. Using

$$\frac{d}{dy}\,\mathrm{Li}_2(-y) = -\frac{\ln(1+y)}{y}, \qquad \frac{d}{dy}\left[\frac{\ln(1+y)}{y}\right] = \frac{1}{y(1+y)} - \frac{\ln(1+y)}{y^2},$$

an antiderivative of half the displayed integrand is

$$-\frac{1}{2}\,\mathrm{Li}_2(-y) - \frac{\ln(1+y)}{y} - \ln(1+y).$$

Its value tends to -1 as $y \to 0$. Thus the exact result within the approximate spectrum is

$$\mathcal{N}(\gamma) = 1 - \frac{1}{2}\,\mathrm{Li}_2(-\gamma^2) - (1+\gamma^{-2})\ln(1+\gamma^2).$$

To expand it at large γ, use the real dilogarithm identity

$$\mathrm{Li}_2(-x) + \mathrm{Li}_2(-1/x) = -\frac{\pi^2}{6} - \frac{1}{2}\ln^2 x, \qquad x > 0.$$

This follows by differentiation; the integration constant is fixed at $x = 1$ by $\mathrm{Li}_2(-1) = -\pi^2/12$. With $r = \gamma^{-2}$, it gives

$$\begin{aligned}\mathcal{N}(\gamma) &= (\ln\gamma - 1)^2 + \frac{\pi^2}{12} + \frac{1}{2}\,\mathrm{Li}_2(-r) - 2r\ln\gamma - (1+r)\ln(1+r) \\ &= (\ln\gamma - 1)^2 + \frac{\pi^2}{12} - \frac{2\ln\gamma + 3/2}{\gamma^2} + O(\gamma^{-4}).\end{aligned}$$

Therefore

$$\boxed{N_\gamma \simeq \frac{z^2e^2}{\pi\hbar c}\left[(\ln\gamma - 1)^2 + \frac{\pi^2}{12}\right].}$$

The marked remainder in the question records the logarithm accompanying the first omitted inverse-square term.

(b) Equation (13.87) gives $I_{\mathrm{all}} = z^2\alpha\hbar\omega_p\gamma/3$. If the energy below ω_p is neglected consistently at leading order in $1/\gamma$, division by part (a) gives

$$\boxed{\langle\hbar\omega\rangle_{\mathrm{leading}} = \frac{\pi\gamma\hbar\omega_p}{3[(\ln\gamma - 1)^2 + \pi^2/12]}.}$$

For $\hbar\omega_p = 0.020$ keV the numerical values are

γ	10^3	10^4	10^5
N_γ/z^2	0.0829805	0.158491	0.258632
$\langle\hbar\omega\rangle_{\mathrm{leading}}$ (keV)	0.58627	3.06951	18.81012

There is a small distinction between this conventional use of the total energy and the mean energy of precisely the photons counted in part (a). For the latter the numerator must have the same cutoff. To quantify it, integration by parts gives

$$J_0(u) = \int_0^u f(\nu)d\nu = \left(u + \frac{2u^3}{3}\right)\ln(1+u^{-2}) - \frac{2u}{3} + \frac{2}{3}\arctan u.$$

Indeed the logarithmic derivative is $-2/[u(1+u^2)]$, and the remaining rational integral is $2\int(1+2u^2/3)/(1+u^2)\,du - 2u$; it gives the last two terms. The limit $J_0(\infty) = \pi/3$ also checks (13.87). Therefore

$$\langle\hbar\omega\rangle_{\omega>\omega_p} = \gamma\hbar\omega_p\,\frac{\pi/3 - J_0(1/\gamma)}{\mathcal{N}(\gamma)} = \begin{cases} 0.57854\text{ keV}, & \gamma = 10^3, \\ 3.06411\text{ keV}, & \gamma = 10^4, \\ 18.80598\text{ keV}, & \gamma = 10^5. \end{cases}$$

Both tables use the supplied high-frequency model down to its nominal cutoff. The second values are the internally consistent cutoff means; the first are the leading large-γ answer obtained directly from (13.87). Their difference is of relative order $\ln\gamma/\gamma$, so it is larger than the inverse-square correction to photon number alone.

13.16 A highly relativistic neutral particle of mass m possessing a magnetic moment μ parallel to its direction of motion emits transition radiation as it crosses at right angles a plane interface from vacuum into a dielectric medium characterized at high frequencies by a plasma frequency ω_p. The magnetic moment μ is defined in the particle's rest frame. [The particle could be a neutron or, of more potential interest, a neutrino with a small mass.]

(a) Show that the intensity of transition radiation is given by (13.79), provided the electric field of the incident particle $\mathbf{E}_i$, is given by $(\beta\mu/\gamma ze)$ times the partial derivative in the z direction of E_ρ in (13.80). Note that the electric field actually points azimuthally, but in the leading ultrarelativistic, small-angle approximation this affects only the polarization of the radiation, not its intensity.

(b) Show that in the combined limit of $\gamma \gg 1$ and $\omega \gg \omega_p$, the intensity distributions in angle and frequency are given by (13.84) and (13.85), each multiplied by $(\mu\omega/ze\gamma c)^2$.

(c) By expressing μ in units of the Bohr magneton $\mu_\mathrm{B} = e\hbar/2m_ec$ and the plasma frequency in atomic units ($\hbar\omega_0 = e^2/a_0 = 27.2$ eV), show that the ratio of frequency distributions of transition radiation emitted by the magnetic moment to that emitted by an electron with the same speed is

$$\frac{dI_\mu(\nu)}{dI_e(\nu)} = \frac{\alpha^4}{4}\left(\frac{\mu}{\mu_\mathrm{B}}\right)^2\left(\frac{\hbar\omega_p}{\hbar\omega_0}\right)^2\cdot\nu^2$$

where $\alpha = 1/137$ is the fine structure constant and $\nu = \omega/\gamma\omega_p$ is the dimensionless frequency variable.

(d) Calculate the total energy of transition radiation, imposing conservation of energy, that is, $\nu \le \nu_\mathrm{max} = mc^2/\hbar\omega_p$. [This constraint will give only a crude estimate of the energy in the quantum regime where $\nu_\mathrm{max} < 1$ because the derivation is classical throughout.] Show that the ratio of total energies for the magnetic moment and an electron of the same speed can be written as

$$\frac{I_\mu}{I_e} = \frac{\alpha^4}{20}\left(\frac{\mu}{\mu_\mathrm{B}}\right)^2\left(\frac{\hbar\omega_p}{\hbar\omega_0}\right)^2\cdot G(\nu_\mathrm{max})$$

where $G \approx 1$ for $\nu_\mathrm{max} \gg 1$ and $G \approx (10\nu_\mathrm{max}^3/\pi)\cdot[\ln(1/\nu_\mathrm{max}) - 2/3]$ for $\nu_\mathrm{max} \ll 1$. For fixed particle energy and magnetic moment, how does the actual amount of radiated energy vary with the particle's mass for very small mass?

Hint: the integrals of Section 2.7 of Gradshteyn and Ryzhik may be of use, although integration by parts is effective.

Solution.

(a) In the particle rest frame, with $\boldsymbol{\mu} = \mu\hat{\mathbf{z}}$,

$$\mathbf{E}' = 0, \qquad \mathbf{B}' = \frac{3\mathbf{r}'(\boldsymbol{\mu}\cdot\mathbf{r}')}{r'^5} - \frac{\boldsymbol{\mu}}{r'^3}.$$

The component required for a boost parallel to the moment is $B'_\rho = 3\mu\rho z'/(\rho^2 + z'^2)^{5/2}$. At a fixed laboratory event $z' = \gamma(z - vt)$ and $\rho' = \rho$. The field transformation gives $\mathbf{E}_\perp = -\gamma\boldsymbol{\beta}\times\mathbf{B}'_\perp$, hence

$$E_{\phi,\mu}(\rho, z, t) = -\frac{3\beta\gamma^2\mu\rho(z - vt)}{[\rho^2 + \gamma^2(z - vt)^2]^{5/2}}.$$

For comparison, the radial field of charge $q = ze$ at the same speed is

$$E_{\rho,q} = \frac{q\gamma\rho}{[\rho^2 + \gamma^2(z - vt)^2]^{3/2}}, \qquad \partial_z E_{\rho,q} = -\frac{3q\gamma^3\rho(z - vt)}{[\rho^2 + \gamma^2(z - vt)^2]^{5/2}}.$$

Thus the asserted relation, including its sign, is

$$\boxed{E_{\phi,\mu} = \frac{\beta\mu}{\gamma ze}\,\partial_z E_{\rho,q}.}$$

The induced polarization and its radiation are still calculated by (13.76)–(13.79), now with this azimuthal incident electric field. Axisymmetry makes the radiation polarized perpendicular to the plane containing the trajectory and observer. Charge radiation is polarized in that plane. Equality of their angular shapes after a scalar multiplication is the leading ultrarelativistic, small-angle result, as established next.

(b) A Fourier component of either driving field contains $e^{i\omega z/v}$. Therefore

$$E_{\phi,\mu}(\omega) = \frac{\beta\mu}{\gamma ze}\frac{i\omega}{v}E_{\rho,q}(\omega) = \frac{i\mu\omega}{ze\gamma c}E_{\rho,q}(\omega).$$

The common longitudinal radiation integral is unchanged. To check the polarization projection explicitly, observe in the x–z plane, put $h = k\sin\theta$ and $\ell = \omega/(\gamma v)$, and write

$$E_{\rho,q} = CK_1(\ell\rho)e^{i\omega z/v}, \qquad E_{z,q} = -\frac{iC}{\gamma}K_0(\ell\rho)e^{i\omega z/v}, \quad C = \frac{ze\omega}{\gamma v^2}\sqrt{\frac{2}{\pi}}.$$

After removing the common z phase, the transverse integrals are

$$T_\rho = -2\pi iC\int_0^\infty \rho K_1(\ell\rho)J_1(h\rho)d\rho = -\frac{2\pi iCh}{\ell(\ell^2 + h^2)},$$

$$T_z = -\frac{2\pi iC}{\gamma}\int_0^\infty \rho K_0(\ell\rho)J_0(h\rho)d\rho = -\frac{2\pi iC}{\gamma(\ell^2 + h^2)}.$$

The second Bessel integral is the radial transform used in (13.81); the first follows from integrating $K_1(\ell\rho) = -\ell^{-1}\partial_\rho K_0(\ell\rho)$ by parts. The charge's transverse radiation component is

$$T_q = \cos\theta\, T_\rho - \sin\theta\, T_z = T_\rho\left(\cos\theta - \frac{1}{\gamma^2\beta n}\right).$$

For the azimuthal field the other polarization has component $T_\mu = i\mu\omega T_\rho/(ze\gamma c)$. In the limit $\gamma \gg 1$, $\theta \ll 1$, $\omega \gg \omega_p$, the parenthesis in T_q tends to one. Hence the amplitudes differ only by $i\mu\omega/(ze\gamma c)$ to the retained order, and

$$\boxed{\left(\frac{d^2I}{d\omega\, d\Omega}\right)_\mu = \left(\frac{\mu\omega}{ze\gamma c}\right)^2 \left(\frac{d^2I}{d\omega\, d\Omega}\right)_q.}$$

The multiplier does not depend on angle, so the same factor multiplies the frequency spectrum (13.85). At finite angle or finite γ the polarization projection above explains the higher-order difference; no exact equality outside the stated approximation is needed.

(c) Set $z = 1$ for comparison with an electron and use $\omega = \gamma\omega_p\nu$. Then

$$\left(\frac{\mu\omega}{e\gamma c}\right)^2 = \left(\frac{\mu}{\mu_\mathrm{B}}\right)^2 \left(\frac{\hbar\omega_p}{2m_e c^2}\right)^2 \nu^2.$$

The atomic relations $a_0 = \hbar/(\alpha m_e c)$ and $e^2 = \alpha\hbar c$ give

$$\hbar\omega_0 = \frac{e^2}{a_0} = \alpha^2 m_e c^2.$$

Thus

$$\boxed{\frac{dI_\mu}{dI_e} = \frac{\alpha^4}{4}\left(\frac{\mu}{\mu_\mathrm{B}}\right)^2 \left(\frac{\hbar\omega_p}{\hbar\omega_0}\right)^2 \nu^2.}$$

The incident neutral-particle mass is m; the electron mass m_e appearing in μ_B and atomic units is a separate quantity.

(d) A photon cannot carry more than the particle energy. For $\gamma \gg 1$, $\hbar\omega \le \gamma mc^2$ gives $u \equiv \nu_\mathrm{max} = mc^2/(\hbar\omega_p)$ as in the question. Using the function $f(\nu)$ defined in Problem 13.15, the cutoff energy is

$$I_\mu(u) = \frac{\mu^2\gamma\omega_p^3}{\pi c^3} J_2(u), \qquad J_2(u) = \int_0^u \nu^2 f(\nu)\, d\nu.$$

Integrate the logarithmic term by parts with

$$P(\nu) = \frac{\nu^3}{3} + \frac{2\nu^5}{5}, \qquad \frac{d}{d\nu}\ln(1+\nu^{-2}) = -\frac{2}{\nu(1+\nu^2)}.$$

This gives

$$J_2(u) = P(u)\ln(1+u^{-2}) + 2\int_0^u \frac{\nu^2/3 + 2\nu^4/5}{1+\nu^2}\, d\nu - \frac{2u^3}{3}.$$

The rational function has the explicit division

$$\frac{\nu^2/3 + 2\nu^4/5}{1+\nu^2} = \frac{2\nu^2}{5} - \frac{1}{15} + \frac{1}{15(1+\nu^2)}.$$

Consequently

$$\boxed{J_2(u) = \left(\frac{u^3}{3} + \frac{2u^5}{5}\right)\ln(1+u^{-2}) - \frac{2u^3}{5} - \frac{2u}{15} + \frac{2}{15}\arctan u.}$$

Its lower-limit value is zero. At large u the polynomial terms cancel against the first terms of the logarithm and $J_2(\infty) = \pi/15$.

For an electron of the same speed, (13.87) gives $I_e = e^2\gamma\omega_p/(3c)$. Division yields

$$\frac{I_\mu(u)}{I_e} = \frac{3\mu^2\omega_p^2}{\pi e^2 c^2} J_2(u) = \frac{\mu^2\omega_p^2}{5e^2c^2}\, G(u), \qquad \boxed{G(u) = \frac{15}{\pi} J_2(u).}$$

Using the conversion in part (c),

$$\boxed{\frac{I_\mu}{I_e} = \frac{\alpha^4}{20}\left(\frac{\mu}{\mu_{\rm B}}\right)^2 \left(\frac{\hbar\omega_p}{\hbar\omega_0}\right)^2 G(u).}$$

The reference electron energy here is the untruncated classical result (13.87), consistent with the question's normalization.

For $u \gg 1$, $f(u) = 1/(6u^4) + O(u^{-6})$, so

$$G(u) = 1 - \frac{15}{\pi}\int_u^\infty \nu^2 f(\nu) d\nu = 1 - \frac{5}{2\pi u} + O(u^{-3}) \longrightarrow 1.$$

For $u \ll 1$, $f(\nu) = 2\ln(1/\nu) - 2 + O(\nu^2 \ln(1/\nu))$. Direct integration gives

$$\begin{aligned} J_2(u) &= 2\left[\frac{\nu^3}{3}\ln\frac{1}{\nu} + \frac{\nu^3}{9}\right]_0^u - \frac{2u^3}{3} + O(u^5\ln(1/u)) \\ &= \frac{2u^3}{3}\left[\ln\frac{1}{u} - \frac{2}{3}\right] + O(u^5\ln(1/u)), \end{aligned}$$

and hence

$$\boxed{G(u) \simeq \frac{10u^3}{\pi}\left[\ln\frac{1}{u} - \frac{2}{3}\right] \quad (u \ll 1).}$$

Finally, hold the total particle energy $\mathcal{E} = \gamma mc^2$ and μ fixed. Then $\gamma \propto m^{-1}$, $u \propto m$, and

$$I_\mu \;\propto\; \gamma J_2(u) \;\propto\; m^2\left[\ln\frac{\hbar\omega_p}{mc^2} - \frac{2}{3}\right] \longrightarrow 0 \qquad (m \to 0).$$

The growth of γ therefore does not produce a divergence once the allowed photon energy is imposed. As stipulated in the question, this mass dependence is only the classical spectrum with a kinematic cutoff; quantum recoil and the radiation matrix element require a quantum calculation when $u < 1$.